Tong Head VCD (**V**ariable **C**enter **D**istance)

The VCD was originally designed to improve ware handling and increase percent pack by slowing the push-out motion and associated conveyor belt speed. This was achieved by reducing ware spacing using the VCD Tong Head.

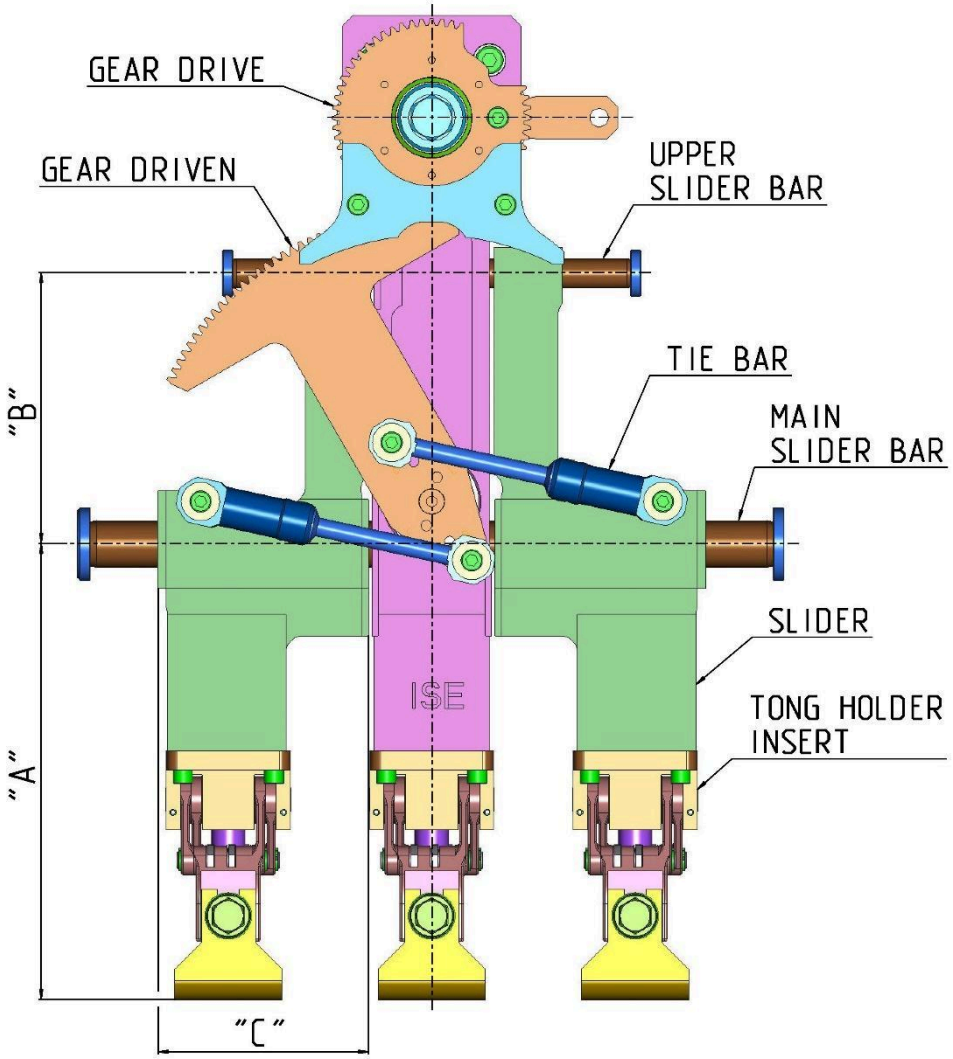
ISE's VCD Tong Head advantages:

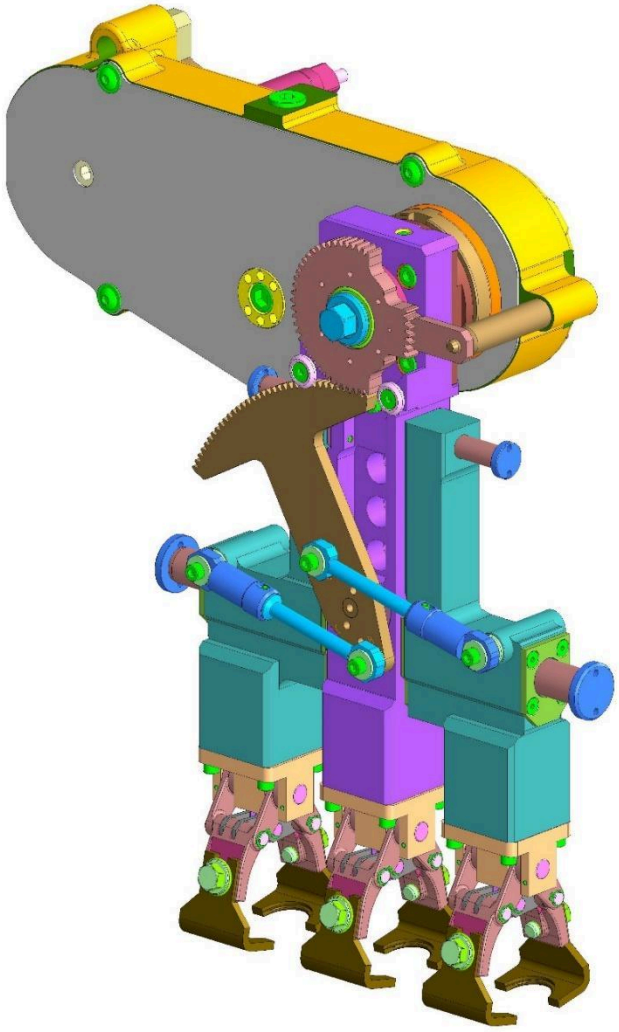
- Smooth container transport to a conveying chain during high-speed production and when producing containers with small diameters
- Slower push-out motion due to the reduced spacing between containers
- Fewer ware handling losses
- More stable production, less bottle contacts
- Fewer crashes on the conveyor belt
- Less traffic jams at the hot end coating hood
- Smooth entry into the ware transfer
- The VCD is an excellent alternative to the very complex dual-row ware handling solution

- ✔ The VCD Tong Head by ISE clearly exceeds current standards regarding robustness and grabbing precision
- ✔ The VCD from ISE has no problems with wear and the associated high maintenance costs.
- ✔ All parts can be easily replaced individually to keep maintenance costs low over a long period of time.
- ✔ Universally applicable, for mounting on ISE, Emhart and Bottero type Take-Out Arms. Mounting on other Take-Out Arms on request.
- ✔ The distance between containers on the conveyor belt is industry standard (other distances on request)
- ✔ Existing Tong Holder Inserts can be used.
- ✔ Available for NIS, IS, AIS, E-IS and ISS forming machines, center distances TG5", QG95mm, TG4-1/4", DG6-1/4", TG85mm, TG3-11/32", TG3-1/2". Emhart and Bottero machines. Others on request

Pos	Features	Benefits
1.	Designed in such a way that any wear on any parts of the VCD does not affect the accuracy of the gripping position over the molds or cause air leakage and loss of the tongs closing air pressure	Reduced critical defects while grabbing the container Maintenance can be planned without urgency Significantly extended service life and greatly reduced maintenance costs Reduced risk for grabbing failure
2.	Large diameters of slider bars increase resistance against bending by > 30%	Increases rigidity and reduces maintenance costs
3.	3x greater load capacity of sliders guiding system	Significantly extends the service life and reduces maintenance costs
4.	Main slider bar is close to tongs (Dimension “A”, see below)	Improves positioning accuracy of tongs, reduces critical defects
5.	Large distance between main slider bar and upper slider bar (Dimension “B”)	Improves positioning accuracy of tongs, reduces vibration and critical defects
6.	Long guiding of sliders on main slider bar (Dimension “C”)	Reduces load on guiding system, reduces wear and improves precision of grabbing position
7.	Guiding system can be greased through grease nipples but guiding system can also run dry	Grease extends the service life and reduces maintenance costs compared with dry running guiding
8.	Metallic scrapers on each side of sliders. Robust guiding system	Prevents inflow of dirt, extends the service life of the linear guide
9.	Large gears. Increased diameter of gears	Extends the service life

Pos	Features	Benefits
10.	Large ball bearing on the drive gear is filled with high temperature grease and sealed. The load capacity has been doubled	Lubricated for life. Extends the service life and reduces maintenance costs.
11.	The VCD can be used as a rigid Tong Head with the same grabbing accuracy as the rigid Tong Head. The sliders are therefore locked at mold distance	Reduce equipment costs. No need for an additional rigid Tong Head. Reduced number of tong holders needed if rigid tong head is not used.
12.	One-Screw only to mount onto Take-Out Arm	Fast and easy tong head replacement
13.	Very robust premium design, built to last	Environmentally friendly thanks to its very long service life
14.	Option: Self-release screw to remove VCD from Take-Out Arm	“Fast and soft” tong head replacement “without hammer”
15.	Option: A protection plate is available for most of the VCDs	The risk of damage to the VCD due to collisions with the Neck-Ring holder arms is significantly reduced


GEAR DRIVE
GEAR DRIVEN
UPPER SLIDER BAR
TIE BAR
MAIN SLIDER BAR
SLIDER
TONG HOLDER INSERT
"B"
"A"
"C"
ISE



Additional advantages of ISE's Tong Head VCD TG4-1/4":

- The risk of damage to the VCD due to collisions with the Neck-Ring holder arms is significantly reduced by using an optionally available protection plate.
- A small conversion kit enables to use the VCD TG4-1/4" also on DG6-1/4" machines.
- Available for Emhart and Bottero type take out arms.
- Different adapter plates allow to switch between Emhart and Bottero type interfaces.
- The VCD can be used as a rigid Tong Head with the same grabbing accuracy as the rigid Tong Head in TG and DG configuration.
- The variable center distances on conveyor belt and the conveyor speed can be set as follows, depending on container diameter:

Machine center distance			
TG4-1/4"		DG6-1/4"	
Distance of bottle centers on conveyor belt	Belt advance per sect. cycle	Distance of bottle centers on conveyor belt	Belt advance per sect. cycle
107.95 (4-1/4")	333.38 (13-1/8")	158.75 (6-1/4")	333.38 (13-1/8")
88.9 (3-1/2")	266.7 (10-1/2")	133.3 (5-1/4")	266.7 (10-1/2")
74.2 (2-59/64")	222.3 (8-3/4")	111.1 (4-3/8")	222.3 (8-3/4")
66.7 (2-5/8")	200 (7-7/8")		
59.5 (2-11/32")	177.8 (7")		

Additional advantages of ISE's Tong Head VCD TG85mm, TG3-11/32" and TG3-1/2":

- The risk of damage to the VCD due to collisions with the Neck-Ring holder arms is significantly reduced by using an optionally available protection plate.
- Small conversion kits can be used to convert the VCD from Emhart's center distance TG85mm to Bottero's TG3-11/32" or TG3-1/2"
- Different adapter plates on the VCD allow to switch between Emhart and Bottero type interface.
- The VCD can be used as a rigid Tong Head with the same grabbing accuracy as the rigid Tong Head.
- The variable center distances on conveyor belt and the conveyor speed can be set as follows, depending on container diameter:

Machine center distance					
TG85mm Emhart		TG3-11/32" Bottero		TG3-1/2" Bottero	
Distance of bottle centers on conveyor belt	Belt advance per sect. cycle	Distance of bottle centers on conveyor belt	Belt advance per sect. cycle	Distance of bottle centers on conveyor belt	Belt advance per sect. cycle
85 (3-1/2")	266.7 (10-1/2")	84.9 (3-11/32")	266.7 (10-1/2")	88.9 (3-1/2")	266.7 (10-1/2")
76.2 (3")	228.6 (9")	76.2 (3")	228.6 (9")	76.2 (3")	228.6 (9")
59.5 (2-11/32")	177.8 (7")	59.5 (2-11/32")	177.8 (7")	59.5 (2-11/32")	177.8 (7")

Additional advantages of ISE's Tong Head VCD TG5" for NIS and ISS:

- Available for NIS and ISS machines
- The variable center distances on conveyor belt and the conveyor speed can be set as follows, depending on container diameter:

Machine center distance	
TG5"	
Distance of bottle centers on conveyor belt	Belt advance per sect. cycle
127 (5")	381 (15")
119 (4-11/16")	357.18 (14-1/16")
107.15 (4-7/32")	321.5 (12-21/32")
95.25 (3-3/4")	285.75 (11-1/4")

Additional advantages of ISE's Tong Head VCD QG95mm for NIS:

- The variable center distances on conveyor belt and the conveyor speed can be set as follows, depending on container diameter:

Machine center distance	
QG95mm	
Distance of bottle centers on conveyor belt	Belt advance per sect. cycle
95 (3-3/4")	381 (15")
76.2 (3")	304.8 (12")
59.5 (2-11/32")	238.13 (9-3/8")

List of VCD Tong Heads

Center Distance	VCD Number	Machine Type	Remarks
TG4-1/4“ & DG6-1/4“	1-19-110	Emhart IS & AIS Emhart type Take-Out arm	
TG4-1/4“ & DG6-1/4“	1-19-109	Bottero IS & E-IS Bottero A57 Take-Out arm	
TG5“	1-19-086	Emhart NIS Emhart type Take-Out arm	
QG95mm	1-19-113	Emhart NIS Emhart type Take-Out arm	
TG5“	1-19-095	Sklostroj ISS Sklostroj Take-Out arm	
TG85mm	1-19-093	Emhart IS Emhart type Take-Out arm	
TG3-11/32“	1-19-112	Bottero IS Bottero A57 Take-Out arm	
TG3-1/2“	1-19-111	Bottero E-IS Bottero A57 Take-Out arm	